

01 GROUNDED TRAINING SAMPLE

C
source evidence

Q
Spanish question

A
reference answer

$$s_i = [c_i \ q_i \ a_i]$$

02 QLoRA OPTIMIZATION

Base LLM
4-bit NF4 · frozen

W_0
NF4

0101
1100
0011

LoRA adapters
all-linear · trainable

A
 $r \times k$

B
 $d \times r$

$$r = 16 \cdot \alpha = 16$$

$$\min_{A,B} - \sum_t \log p_{W_0,A,B}(a_{i,t} \mid c_i, q_i, a_{i,1:t-1})$$

03

Grounded QLoRA checkpoint

base model reference + compact PEFT adapter

04 CLOSED-BOOK SFT ABLATION

Q only
context removed

$$s_i^{c^B} = [q_i \ a_i]$$

Only A and B receive optimizer updates; the quantized backbone remains frozen.